1. June 15/4H/Q7 a & d

(a) Expand $6(4 - 3y)$ $24 - 18y$

(1)

(d) Expand and simplify $(y + 10)(y - 2)$

$y^2 - 2y + 10y - 20 = y^2 + 8y - 20$

(2)

2. June 15/4HR/Q9 a

(a) Expand $5(2p - 3)$ $10p - 15$

(1)

3. Jan 16/3H/Q4 c

Expand and simplify $(x + 5)(x - 1)$

$x^2 - x + 5x - 5 = x^2 + 4x - 5$

(2)

4. Jan 16/3HR/Q2 b

(b) Expand and simplify $(t - 4)(t + 5)$ $t^2 + 5t - 4t - 20 = t^2 + t - 20$

(2)

5. Jan 16/4H/Q9 a

Expand $x(x + 2)$ $x^2 + 2x$

(1)

6. Jan 16/4HR/Q12

Simplify fully $(2x + 3)^2 - (2x - 3)^2$

$4x^2 + 12x + 9 - (4x^2 - 12x + 9)$

$= 4x^2 + 12x + 9 - 4x^2 + 12x - 9$

$= 24x$

(3)

Scanned with CamScanner

7. June 16/3H/Q7 b

Expand and simplify $(x - 9)(x + 2)$

$x^2 + 2x - 9x - 18 = x^2 - 7x - 18$ (2)

8. June 16/3HR/Q12 c

Expand and simplify $(2y + 3)(4y - 1)$

$8y^2 - 2y + 12y - 3 = 8y^2 + 10y - 3$ (2)

9. June 16/4H/Q5 d

Expand and simplify $4(2x + 3) + 2(x + 5)$

$8x + 12 + 2x + 10 = 10x + 22$ (2)

10. June 16/4HR/Q2 b

Expand and simplify $3(2y + 2) + 2(y - 4)$

$6y + 6 + 2y - 8 = 8y - 2$ (2)

11. June 16/4HR/ Q14 c

Expand and simplify $4x(x + 3) - (2x - 3)^2$

$4x^2 + 12x - (4x^2 - 12x + 9) =$

$4x^2 + 12x - 4x^2 + 12x - 9 = 24x - 9$ (3)

12. Jan 17/3H/Q5 b

Expand and simplify $4x(x + 3) - (2x - 3)^2$

$4x^2 + 12x - (4x^2 - 12x + 9)$

$= 4x^2 + 12x - 4x^2 + 12x - 9$

$= 24x - 9$ (3)

Scanned with CamScanner

13. Jan 17/3H/Q9

Simplify $(2x + 3)^2 - (2x + 3)(x - 5)$

Give your answer in the form $ax^2 + bx + c$

$4x^2+12x+9-(2x^2+10x+3x-15)$

$= x^2+12x+9-(2x^2-7x-15)$ (3) $= x^2+12x+9-2x^2+7x+15$

$= 3x^2+19x+24$

14. Jan 17/3HR/Q11 b

Expand and simplify $(k + 9)(k - 5)$ $k^2-5k+9k-45$

$= k^2+4k-45$ (2)

15. Jan 17/4HR/Q7 a

Expand $3(4p + 5)$ $12p+15$ (1)

16. Jan 18/4H/Q10 b & d

Expand $p(2p - 3)$ $2p^2-3p$ (1)

17. Jan 18/4HR/Q7 c

Expand and simplify $(x + 3)(x - 6)$

$x^2-6x+3x-18=x^2-3x-18$ (2)

18. June 18/1H/Q11 a

Expand and simplify $(2x - 1)(x + 3)(x - 5)$

$(2x-1)(x^2-5x+3x-15)=(2x-1)(x^2-2x-15)$

$2x^3-10x^2$ $= 2x^3-4x^2-30x-x^2+2x+15$ (3)

$= 2x^3-5x^2-28x+15$

Scanned with CamScanner

19. June 18/1HR/Q14 b

Expand and simplify $(3x - 4y)(x + 3y)$

$3x^2 + 9xy - 4yx - 12y^2 = 3x^2 + 5xy - 12y^2$ (2)

20. Jan 19/1H/Q1 b

Expand and simplify $(e + 3)(e - 5)$

$e^2 - 5e + 3e - 15 = e^2 - 2e - 15$ (2)

21. Jan 19/1HR/Q15 b

Expand and simplify $4n(n - 3)(n + 5)$

$4n^2(n^2 + 5n - 3n - 15) = 4n^3 + 8n^2 - 60n$ (2)

22. June 19/1H/Q3 c

Expand and simplify $(x + 9)(x - 2)$

$x^2 - 2x + 9x - 18 = x^2 + 7x - 18$ (2)

23. June 19/1HR/Q18 a

Expand and simplify $(x + 2)(2x + 3)(x - 7)$
Show your working clearly.

$(x+2)(2x^2 - 14x + 3x - 21)$
$= (x+2)(2x^2 - 11x - 21) = 2x^3 - 11x^2$

24. Jan 20/1H/Q7 a

Expand and simplify $(m - 8)(m + 5)$

$m^2 - 3m - 40$ (2)

25. Jan 20/1HR/Q3 c

Expand and simplify $(x + 2)(x - 5)$

$x^2 - 3x - 10$ (2)

Scanned with CamScanner

26. June 20/1HR/Q14

Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

$20x^2 - 31x^2 - 8x - 18$

(3)

27. Jan 21/1H/Q12 b

Expand and simplify $2x(x - 5)(x - 3)$

$2x^3 - 16x^2 + 30x$

(3)

28. Jan 21/1HR/Q16 a

Expand and simplify $(x + 4)(x - 2)(x + 1)$

$x^3 + 3x^2 - 6x - 8$

(3)

29. Jan 21/2H/Q5 a

Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

$2x^2 + 29x$

(2)

30. June 21/1H/Q13

Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

$24x^3 - 28x^2 - 12x$

(3)

31. Nov 21/2H/Q12 c

Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly.

$8x^3 - 28x^2 - 60x$

(2)

Scanned with CamScanner

32. Jan 22/1H/Q13 a

Expand and simplify $5x(x + 2)(3x - 4)$

$15x^3 + 10x^2 - 40x$

(2)

33. Jan 22/1HR/Q12 a

Expand and simplify $n(n - 4)(3n + 5)$

$3n^3 - 7n^2 - 20n$

(2)

34. Jan 23/1H/Q13

Expand and simplify $3x(2x - 5)^2$
Show clear algebraic working.

$12x^3 - 60x^2 + 75x$

(3)

35. Jan 23/2HR/Q12 c

Expand and simplify $5x(3x + 4)(2x - 1)$

$30x^3 + 25x^2 - 20x$ (2)

36. June 23/2H/Q17 a

Expand and simplify $(x + 6)(3x - 2)(x + 6)$

$3x^3 + 34x^2 + 84x - 72$

(3)

Scanned with CamScanner

37. June 24/1HR/Q 14

(a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$

$-3x^3 - 7x^2 + 22x + 8$

(3)

(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

a^5b^2

(3)

Scanned with CamScanner